

★ SPECIALISTS IN COMPRESSOR SPARES

Engineered PTFE & PEEK for the toughest environments.

Compressor spares · Rods · Bushes · Seals & Rings · Bellows · Gaskets · Bearings · Custom components



One name for six decades of PTFE & PEEK engineering.

Since 1963, **Corrosion Engineers** has moulded and machined PTFE, filled-PTFE and PEEK components in Ahmedabad for India's most demanding industries — railways, defence, atomic energy, power, petrochemicals and heavy engineering.

Corvex is the new face of Corrosion Engineers. Same materials, same craftsmanship, same people — now one bold name on every rod, ring, seal and bellows we ship.



Wherever parts slide, rotate or oscillate — or wherever corrosive chemicals, friction or extreme temperatures destroy other materials — there is a job for [Corvex](#).

1963

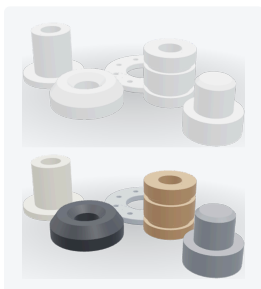
Established in Ahmedabad

28

Industries served

50+

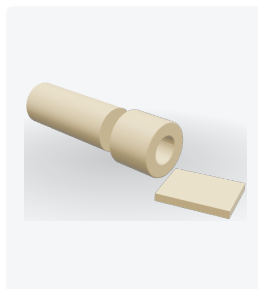
Institutional & PSU customers



PTFE →

Self-lubricating, chemically inert, -268 to +260 °C.

- Virgin & filled grades ›
- Rods, tubes & parts ›
- Sheets & pads ›



PEEK →

Strength and wear resistance for load and heat.

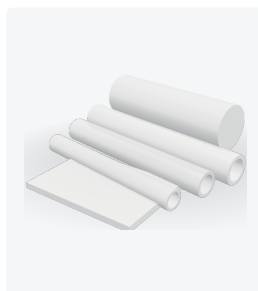
- PEEK grades ›
- Rods & bushes ›
- Machined parts ›



Compressor spares →

Rings and valve plates, to drawing or OEM part number.

- Piston & rider rings ›
- Packing & oil scrapers ›
- Valve plates & discs ›



Other engineering polymers →

Fluoropolymers and tough nylon: rods, sheets, tubes.

- FEP & PFA ›
- PCTFE & PVDF ›
- Nylon ›

Made to drawing

Standard sizes from stock, plus non-standard sizes and filled compositions made to your specification.

Proven in the field

Registered vendor to Indian Railways, BHEL, ONGC, DAE, SAIL and the Ministry of Defence — and supplier to Burckhardt Compression.

Checked before dispatch

Machined to match your tolerances, with QC before delivery — precision-engineered rings in the minimum lead time possible.



Corvex PTFE — the self-lubricating engineering material.

Corvex PTFE markedly improves the performance and service life of wear elements, seals and other key components in hydraulic, pneumatic and mechanical systems. In its virgin state it offers a coefficient of friction lower than graphite, outstanding temperature resistance, and corrosion and chemical resistance equal to or better than glass.

Compounded with inorganic fillers, it gains wear resistance comparable to many soft-metal alloys — superior to most self-lubricating materials. Freedom from stick-slip, excellent plastic memory and the ability to absorb contaminants make it the answer where hostile environments rule out lubricants or conventional parts.

-268 to +260 °C

Service temperature

0.05

Dynamic friction coefficient

55 kV/mm

Dielectric strength

<0.01 %

Water absorption

Why Corvex PTFE is preferred — its key properties

01 Chemical inertness
02 High dielectric strength
03 Heat resistance
04 Toughness & flexibility
05 Low coefficient of friction
06 Noise damping
07 Weather resistance
08 Light weight
09 High impact strength
10 Long life
11 Self-lubricating

The grades we offer — and their properties

Test methods: ASTM D792, D638, D2240, D621, D149

GRADE	DENSITY G/CM ³	TENSILE KGF/CM ²	ELONGATION %	HARDNESS SHORE D	DEFORM. 24 H %	DIELECTRIC KV/MM
 Virgin PTFE Maximum chemical resistance, lowest friction. The all-rounder.	2.1 – 2.2	210 – 375	250 – 400	52 – 58	10	22 – 24
 Chemically modified Denser, lower creep, smoother surface; also electrical grade.	2.15 – 2.2	300 – 325	350 – 450	55 – 62	5	30 – 35
 25% glass Better wear and creep resistance; chemically resistant.	2.22 – 2.25	125 – 200	200 – 300	58 – 63	11	11 – 12
 15% glass + 5% MoS₂ Harder, low-friction surface for dynamic seals and rings.	2.2 – 2.24	150 – 220	220 – 320	60 – 65	13	15 – 16
 25% carbon (or 23% + 2% graphite) Higher compressive strength; good thermal conductivity.	2.0 – 2.2	125 – 200	80 – 150	60 – 65	7	1 – 2
 15% graphite Very low friction and wear; suits dry running.	2.1 – 2.16	150 – 200	150 – 250	60 – 65	8	1 – 2
 40% bronze High wear resistance and heat dissipation for bearings.	3.0 – 3.2	125 – 225	200 – 350	62 – 66	6	Conductive
 40% bronze + 5% MoS₂ As 40% bronze, with a lower-friction surface.	3.0 – 3.2	125 – 225	200 – 350	62 – 66	6	Conductive
 60% bronze Highest load capacity for heavy bearing duty.	3.8 – 4.0	100 – 200	150 – 300	64 – 68	5	Conductive

Also on request: Rulon-type (red) bearing compound for bushes, liners and guides, and Turcite-type low-friction slideway material for guide strips and wear bands. All grades: melting point 335 °C, continuous service to +260 °C. Typical values for guidance only — not design limits. Trade names describe equivalent compound types only.



PTFE rods & tubes.

Extruded and moulded stock shapes in virgin and filled PTFE, ready for machining.
Any size outside these ranges is made to order.



VIRGIN



FILLED

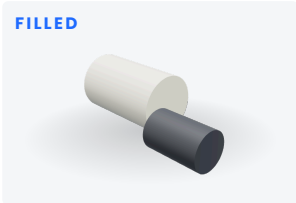
01
Extruded rods

Diameter	3 – 100 mm
Length	1000 & 2000 mm
Made to order	Custom length & diameter

Also in filled compounds — glass, carbon, graphite, bronze, MoS₂.



VIRGIN



FILLED

02
Moulded rods

Diameter 16 – 400 mm	Length 300 mm
Diameter 105 – 300 mm	Length 1000 mm
Made to order	Custom length & diameter

Also in filled compounds — glass, carbon, graphite, bronze, MoS₂.



VIRGIN



FILLED

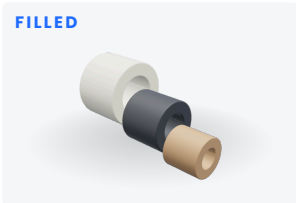
03
Paste-extruded tubes

OUTER DIAMETER (MM)	WALL (MM)
3.2, 4.8, 6.4, 8, 9.5, 11.1, 12.7, 14.3	0.8
3, 4, 6, 8, 10, 12, 14	1.0
6.35, 8, 9.5, 11.1, 12.5, 16, 19, 22.2, 25, 28	1.5

Custom length, OD and ID on request.



VIRGIN

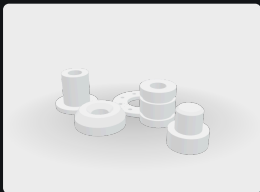


FILLED

04
Moulded tubes & bushes

OD / ID range	12.5 – 1500 mm
Note	OD and ID are co-related
Made to order	Custom length, OD & ID

Also in filled compounds — glass, carbon, graphite, bronze, MoS₂.

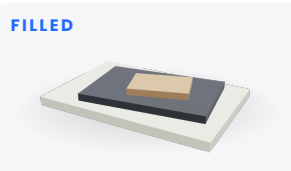
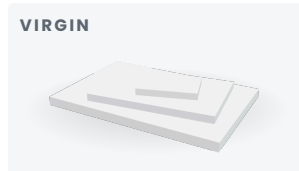



Machined components to drawing

Valve seat rings, wedge rings, plug-valve sleeves, O-rings, washers, bushes, balls, insulators, pipe joints, guides, hydraulic seals — and every PTFE part for non-lubricated compressors — machined from any grade to your sample, specification or drawing.



PTFE sheets, pads & special-purpose grades.



05

Moulded sheets

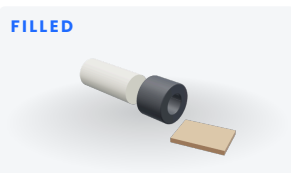
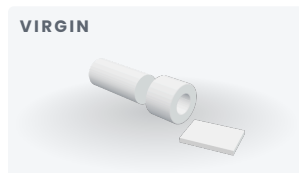
Thickness	3 – 100 mm
Sizes (mm)	300 × 300 up to 1500 × 1500; 1000 × 2000



06

Skived sheets & film

Thickness	0.2 – 6 mm
Width	300, 450, 600, 1000, 1200, 1500, 2000 mm

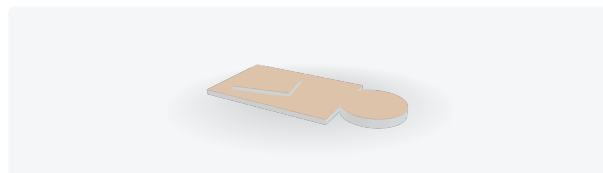


07

Modified PTFE (TFM-type)

Chemically modified PTFE: lower creep, less porosity and smoother surfaces than virgin grade.

Rods	Ø 12 – 400 × 300 mm
Moulded bushes	OD / ID 12.5 – 1500 mm
Skived sheets	300 – 1500 wide · 0.2 – 6 mm
Moulded sheets	300 × 300 – 1500 × 1500 · 8 – 100 mm

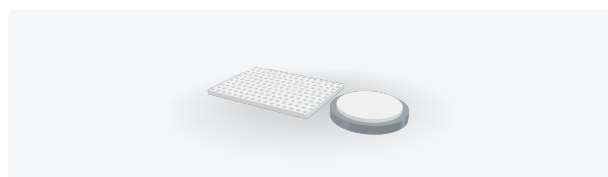


08

Chemically etched (bondable) PTFE

Etched on one or both sides so PTFE bonds to steel, wood, glass and plastics with standard adhesives. Only the surface changes (turns brown).

Forms	Sheets, film, rods, tubes, pads
Uses	Linings for chutes, hoppers, tanks, rollers
Storage	Keep wrapped, away from UV light

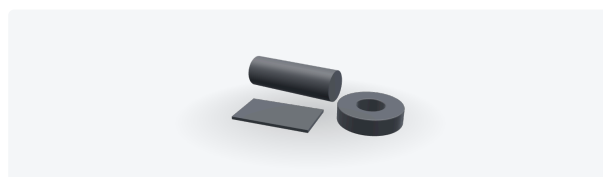


09

Bridge & structural bearing pads

Plain or dimpled PTFE sliding pads between deck and pier — high load capacity, low friction, corrosion- and weather-proof, self-lubricating.

- Multi-span bridges
- Steel & concrete structures
- Pipe supports
- Tank & vessel bases



10

Anti-static PTFE

Conductive PTFE rods, spacers and sheets for lined pipes and fittings where static build-up must be dissipated.

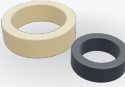
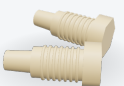
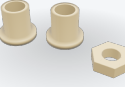
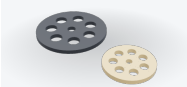
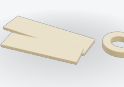
Specific gravity	2.14 – 2.20
Tensile strength	20 – 30 MPa
Hardness	Shore D 55 – 65
Volume / surface resistivity	< 10 ⁶ Ω·cm / < 10 ⁶ Ω



Corvex PEEK — strength where PTFE gives way.

PEEK (polyetheretherketone) keeps its strength, stiffness and shape at temperatures that soften most plastics. Where PTFE would creep or cold-flow under high load and pressure, Corvex PEEK holds its dimensions — making it the material of choice for compressor valve plates, high-pressure rings and precision structural parts.

Where PEEK earns its place Stock in virgin & filled grades — rods, sheets, bushes — and finished parts to drawing

 Compressors Valve plates, poppets, piston & rider rings, back-up rings, pressure breakers	 Oil & gas Back-up & seal rings, valve seats, insulating bushes, connectors	 Pumps Wear & case rings, sleeve bearings, thrust washers
 Valves High-pressure / high-temperature seats, stem seals	 Chemical & process Pump parts, fittings, seals, filter parts	 Electrical Insulators, connectors, coil formers, terminal blocks
 Semiconductor & instruments Wafer-handling & test-socket parts, precision parts	 Food & pharma Valve seats, scraper blades, filling-machine parts	 Automotive & machinery Thrust washers, bushes, gears, seal rings

PEEK grades

Virgin PEEK  Tough, strong and chemically resistant. The standard grade for most parts. Ask for this grade →	Glass-filled PEEK  30% glass for higher stiffness, dimensional stability and heat deflection. Ask for this grade →	Carbon-filled PEEK  30% carbon fibre: highest strength and stiffness, better wear and thermal conductivity. Ask for this grade →	Bearing-grade PEEK  Carbon, graphite and PTFE modified for the lowest friction and wear in dry running. Ask for this grade →
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Typical properties — virgin PEEK

Density	1.30 – 1.32 g/cm³
Tensile strength	95 – 110 N/mm²
Elongation at break	20 – 45 %
Flexural modulus	≈ 4,000 N/mm²
Hardness	Shore D 85 – 88
Melting point	343 °C
Continuous service temperature	up to 250 °C
Heat deflection, 1.8 N/mm²	≈ 152 °C (filled: ≈ 315 °C)
Dielectric strength	≈ 20 kV/mm

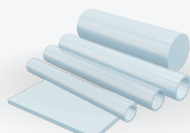
	PTFE	PEEK
Continuous service temperature	260 °C	250 °C
Tensile strength	20 – 35 N/mm²	95 – 110 N/mm²
Creep / cold flow under load	High	Very low
Friction	Lowest of any solid	Low (bearing grades lower)
Chemical resistance	Virtually universal	Excellent (not conc. sulphuric / nitric acid)
Best for	Seals, gaskets, bellows, low-load sliding	High-load rings, valve plates, structural parts



FEP, PFA, PCTFE, PVDF & Nylon.

Standard rods and sheets — call or WhatsApp us for sizes and prices.

ROD · TUBING · SHEET



FEP

−196 to +200 °C

Transparent, non-stick, excellent electrical insulation.

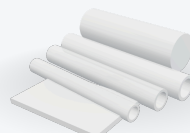
Rods **Ø 6 – 135 mm · L ≤ 200 mm**

Sheets **3 – 60 mm · 120 × 200 to 645 × 560 mm**

Skived sheets **1 – 4 mm · 1000 / 1200 mm wide**

Tubing **ID 1 – 50 mm · wall 0.1 – 4 mm · L ≤ 6 m**

ROD · TUBING · SHEET



PFA

−200 to +260 °C

Melt-processable with PTFE-like heat and chemical resistance; UV-stable.

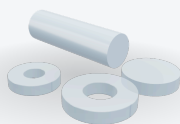
Rods **Ø 5 – 100 mm · L 1000 mm**

Sheets **3 – 60 mm · 120 × 200 to 645 × 560 mm**

Skived sheets **1 – 4 mm · 1000 / 1200 mm wide**

Tubing **OD 2 – 43 mm · wall 0.5 – 2.5 mm**

ROD · SEAT RINGS · DISC



PCTFE

−196 to +120 °C

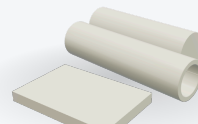
Hard, dimensionally stable, gas-tight — for cryogenic and high-pressure seals.

Rods **Ø 3 – 100 mm · L 1000 mm**

Sheets **3 – 60 mm · 120 × 200 to 500 × 500 mm**

Tubes **OD × ID 34 × 16 to 640 × 610 mm · L 25 – 170 mm**

ROD · PIPE · SHEET



PVDF

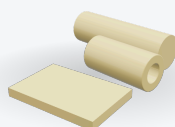
−30 to +150 °C

Strong, abrasion-resistant and flame-retardant — for tank linings, fittings and water treatment.

Rods **Ø 10 – 200 mm · L 1000 mm**

Sheets **2 – 50 mm · 1000 × 2000, 1500 × 3000 mm**

ROD · BUSH · SHEET



Nylon

−40 to +100 °C

Tough, strong and wear-resistant engineering plastic — for bushes, gears, rollers and wear pads.

Rods **Standard diameters, cut to length**

Sheets **Standard thicknesses and sizes**

TYPICAL PROPERTY	UNIT	FEP	PFA	PCTFE	PVDF	NYLON
Density	g/cm ³	2.12 – 2.17	2.12 – 2.17	2.10 – 2.20	1.75 – 1.78	1.13 – 1.15
Tensile strength	MPa	18 – 25	25 – 35	38 – 48	42 – 55	70 – 85
Elongation at break	%	150 – 350	200 – 380	40 – 100	20 – 50	20 – 50
Hardness	Shore D	56	60	85	80	80 – 85
Melting point	°C	275	310	220	166	220 – 260
Dielectric strength	kV/mm	20 – 24	20	20 – 24	10	20 – 25

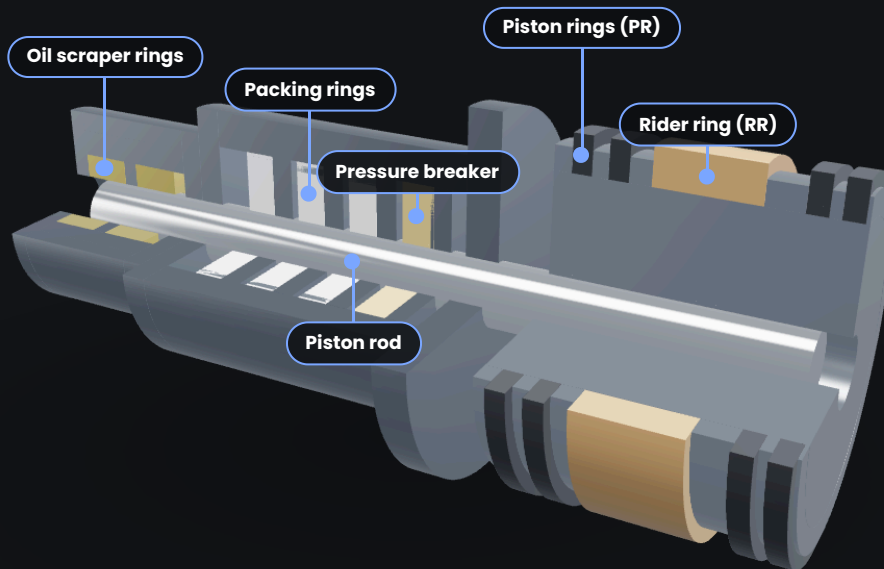
Typical values for guidance only; not to be used as specification limits. Custom sizes on request.



Compressor spares — where Corvex is strongest.

Compressor components are the heart of our business. We make PTFE, filled-PTFE and PEEK wear parts for reciprocating air, gas and refrigeration compressors — for OEMs, and for maintenance teams who need a reliable replacement for any make, made to sample or drawing.

Gas-specific & high-pressure specialists. For gas-specific and high-pressure compressors, we engineer the best solution using our **proprietary filler compositions**, delivering extraordinary efficiency and long service life with minimal wear.



Runs oil-free

Self-lubricating rings end oil carry-over and line contamination.

Protects the cylinder

PTFE absorbs stray particles instead of scoring the liner.

Any make, any size

Machined to your sample, drawing or OEM part number.

7 proprietary grades

In-house compressor compounds, plus 13 generic grades, matched to the gas, pressure and duty.

SUPPLIER TO



Since 2024

REPLACEMENT SPARES FOR

Bharat Pumps & Compressors Ltd (BP&CL)

Chicago Pneumatic

Kirloskar Pneumatic

K.G. Khosla

Ingersoll Rand

Dresser-Rand

Atlas Copco

Brand names belong to their respective owners; Corvex supplies compatible replacement parts.



Compressor oil, with your spares

Lubricating oils for reciprocating and screw compressors, delivered with your ring and valve order. Tell us the make and model and we recommend the grade.

- Reciprocating & screw
- One order, one delivery
- Right grade, first time

Compressor rings leaflet: www.corvex.co.in/Corvex-Compressor-Rings-Leaflet.pdf



Four part families. Thirteen grades.

Send a sample, a drawing or the OEM part number. WhatsApp **+91 73508 08899** with the part name for a quick quote.



Piston rings

Carbon- and graphite-filled PTFE and PEEK rings, in every style, for non-lubricated and lubricated cylinders.

● Carbon ● Graphite
● MoS₂ ● PEEK



Rider rings & wear bands

Bronze- and carbon-filled bands that carry the piston and protect the liner.

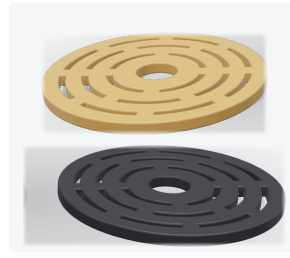
● Bronze ● Carbon



Packing rings & oil scrapers

Segmental rod packing, pressure breakers and bronze oil wiper and scraper rings that keep oil off the rod.

● Carbon ● Graphite
● PEEK ● Bronze



Valve plates & discs

PEEK and filled-PTFE valve plates, discs and seats for suction and delivery valves.

● PEEK ● Carbon ● Glass

13 compressor grades, matched to your application

Grade	Description	Air	Industrial gases	Natural gas	Refinery	Olefins	Alcohols	Chemicals	Refrigeration
CX101	Premium filled-PTFE alloy for the widest range of gases, refrigeration included.	●	●	●	●	●		●	●
CX103	PEEK-based compound for hot service, lubricated or non-lubricated.	●	●	●	●	●	●	●	
CX534	PEEK-based compound for high-pressure, high-temperature lubricated duty.	●	●	●	●	●	●	●	
CX29	Glass-fibre filled PTFE for industrial gases and chemical service.		●					●	
CX52	Field-proven grade for non-lubricated air compressors.	●	●						
CX54	Versatile all-rounder from wet to dry gas, lubricated or non-lubricated.	●	●	●	●	●		●	●
CX60	Filled PTFE for pure gases and oxygen-rich service.		●						
CX79	Stiffer, low-extrusion compound for demanding air duty.	●							
CX121	Low friction with high wear resistance at very low dew points.		●	●	●	●			
CX509	Glass / MoS ₂ filled PTFE with fibre; gentle on soft counter-faces.	●	●	●	●	●	●	●	
CX538	Low-creep PTFE for bone-dry gas (dew point -60 °C or lower), non-lubricated.	●	●	●	●	●			
CX22	Carbon-graphite filled PTFE; lowest wear of the medium-to-high filled grades.	●	●	●	●	●	●	●	●
CX112	Carbon-filled PTFE; dependable value grade for lubricated natural gas.			●					

● Standard ● Performance ● Premium

Typical applications for guidance only. Cross-reference to OEM grades on request.



Engineered for 28 industries.

Aeronautical

- Insulating rings
- Sealing rings
- Spacers

Battery manufacturers

- Ejection tips
- Plungers

Chemical plants

- Compressor components
- Envelope gaskets
- Gaskets
- Sheets
- Valve bellows
- Washers

Civil engineering

- Bearing pads for bridges
- O-rings

Compressor manufacturers & users

- Piston, rider & packing rings
- Valve plates, discs & guides
- Guide, wear & pressure rings
- Oil wiper & scraper rings

Dairy

- Compressor components
- Pump seats for milk pumps

Diesel engines

- Seals in special PTFE compositions

Duplicating machines

- Rollers
- Tapes

Electrical & electronics

- Balls
- Insulating bushes
- Carbon-filled bushes
- O-rings
- Sealing rings
- Sheets

Engineering

- Bearings, bushes & collars (pure & filled)
- O-rings
- Packings
- Rings & rods (pure & filled)
- Sheets
- Washers & gaskets
- V-seals

Fertilizer

- Compressor components
- Pump components
- Seals & seal rings
- Sheets
- Valve components

Hydraulic

- Back-up rings
- O-rings
- Rod seals
- Sealing rings
- U-seals
- V-seals

Instrumentation

- Bushes
- Compressor components
- Gaskets
- O-rings
- Seals & seats
- Valve seats
- V-rings
- Washers

Offshore

- Bearing pads
- Bushes
- Compressor components
- Gaskets
- Packing
- Rods
- Sheets

Oxygen & industrial gas

- Compressor components
- Gaskets
- Gland packing rings
- Guide rings
- Expansion-engine piston rings
- Ring gaskets

Petrochemical

- Bushes
- O-rings
- Packing
- Compressor components

Pharmaceutical

- Bushes
- Compressor parts
- Envelope gaskets
- Stirrer bellows
- Valve bellows

Pulp & paper

- Bushes
- Compressor components

Pumps

- Bearings
- Glass-filled PTFE bushes
- Diaphragms

Railways

- Bearing pads for bogies, locomotives & bridges
- Sheets

Scientific instruments

- Bellow stirrers
- Discs
- Valve bellows

Spare-parts dealers

- PTFE / filled PTFE
- Components for hydraulics, pumps & valves
- Electrical applications
- Tapes

Steel plants

- Compressor components
- Gaskets
- Sheets
- Tape

Sugar

- Bushes
- Compressor components
- Rods
- Sheets

Tools

- Carbon-filled bushes

Tractors

- Hydraulic equipment components
- PTFE-coated O-rings

Tyres

- Belts
- Piston rings for Bagomatic / bomb presses

Valve manufacturers

- Balls
- Bushes
- Gaskets
- O-rings
- Seal rings
- Valve seats
- Washers

Industry not listed?

Tell us the medium, temperature, pressure and motion. We will suggest the grade and make the part to your drawing.

 **+91 73508 08899**



In good company.

For over sixty years, India's leading manufacturers, PSUs and defence establishments have relied on our PTFE components. Registered vendor to Railways, BHEL, ONGC, SAIL, HAL, DAE and the Ministry of Defence.



Supplier of compressor rings since 2024



Customer since 2021

• 28 registered vendors — registration numbers on request

Chemicals, fertilisers & process 5

- Gujarat State Fertilizers Co. Ltd.
- Coromandel Fertilizer Corpn. Ltd.
- Gujarat Heavy Chemicals Ltd.
- Gujarat Alkalies & Chemicals Ltd.
- Krishak Bharati Co-operative Ltd.

Defence, aerospace & atomic 6

- NPOL, Kochi
- Garden Reach Shipbuilders & Engineers Ltd.
- Ministry of Defence
- MoD — Ordnance Factory
- Department of Atomic Energy
- DGQA, Ministry of Defence

Steel & heavy industry 2

- Bhilai Steel Plant
- SAIL — Rourkela Steel Plant

Power & electrical 4

- BHEL
- Hyderabad Batteries Ltd.
- Indian Cable Co. Ltd.
- NTPC

Railways 6

- Western Railway
- Central Railway
- Northern Railway
- South Central Railway
- Integral Coach Factory
- Eastern Railway

Oil, gas & petrochemicals 5

- Hindustan Petroleum (HPCL)
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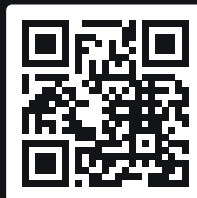
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